

VIGNESH ARUMUGAM

AI Engineer | Generative AI & LLM Systems | Full-Stack Foundations

IMMEDIATE JOINER

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Open to GCC relocation: UAE, Saudi Arabia, Qatar, Kuwait, Bahrain, Oman



PROFESSIONAL SUMMARY

Software Engineer with 1+ year of hands-on experience designing, building, and deploying Generative AI and LLM-powered systems in production environments. Skilled in end-to-end AI pipeline architecture, prompt engineering, Retrieval-Augmented Generation (RAG), AI agents, LangChain, vector databases, and backend API development with Python and FastAPI. Experienced integrating multiple LLM APIs (OpenAI, Google Gemini, Groq) through provider-agnostic, configurable architectures, containerizing AI services with Docker, and deploying to cloud and Kubernetes (Amazon EKS) environments while optimizing latency, cost, and reliability at scale. Strong full-stack foundation (Java, Spring Boot, React.js) with a growing specialization in scalable, production-grade AI applications.

PERSONAL DETAILS

Nationality: Indian

Languages Known: English, Tamil

Availability: Immediate

PROFESSIONAL EXPERIENCE

Software Engineer — Navat Technology, Chennai

Aug 2025 – Present

- Developed and integrated AI/ML solutions using LLMs and Generative AI models into production applications, improving feature responsiveness and output quality.
- Designed end-to-end AI pipeline architecture covering prompt generation, validation, response processing, and error handling for consistent, high-quality outputs.
- Built and maintained AI-powered backend services and RESTful APIs using Python and FastAPI, supporting high-concurrency, low-latency AI workloads.
- Engineered provider-agnostic, configurable architectures to integrate multiple LLM APIs (OpenAI, Google Gemini, Groq) and open-source models, enabling seamless model switching and reliable model serving.
- Implemented prompt engineering strategies, automated quality checks, regeneration logic, and fallback and retry mechanisms to maximize output reliability.
- Built Retrieval-Augmented Generation (RAG) and AI agent workflows using LangChain, embeddings, and vector databases to ground LLM responses in domain data and automate multi-step tasks.
- Worked hands-on with Text-to-Speech (TTS) and multimodal AI technologies, integrating speech and multimodal capabilities into applications.
- Evaluated open-source AI models and local/self-hosted inference solutions (Ollama, Hugging Face, ONNX) to reduce operating costs and improve flexibility.
- Containerized AI microservices using Docker and operated them within Linux-based production environments for consistent, portable deployments.
- Deployed containerized AI services to Amazon Elastic Kubernetes Service (EKS) and integrated releases into CI/CD pipelines for scalable, repeatable deployments.
- Debugged and optimized AI application performance across API failures, memory usage, and model execution bottlenecks, improving system stability.
- Benchmarked response latency, throughput, and output accuracy across LLM providers, and set up logging and monitoring dashboards for real-time visibility.
- Reduced LLM API costs through response caching, request batching, and evaluation of self-hosted open-source models while maintaining low latency.
- Secured REST API endpoints with JWT-based authentication and integrated them with React.js front ends and MySQL-backed services.
- Authored unit and integration tests for AI service endpoints, strengthening output consistency and regression coverage.
- Collaborated cross-functionally with product, backend, and QA teams to scope AI feature requirements and drive successful production launches.
- Followed Agile/Scrum practices with Git/GitHub version control to deliver AI features iteratively and ship reliable releases.

TECHNICAL SKILLS

AI / Generative AI: Large Language Models (LLMs), LLM Engineering, Generative AI, Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain, Vector Databases, Embeddings, AI Agents, Text-to-Speech (TTS), Multimodal AI, Model Integration, Model Serving, Provider Abstraction, Fallback & Retry Handling, Quality Gates

Languages: Python, Java, SQL, TypeScript, JavaScript

Backend Development: FastAPI, REST API Design, Node.js, Spring Boot, Hibernate, JWT Authentication

AI Tools & Platforms: LangChain, OpenAI API, Google Gemini API, Groq API, Ollama, Hugging Face, ONNX, Vector Databases, Open-Source & Local Model Inference

Web Technologies: HTML5, CSS3, React.js, Bootstrap

Databases: MySQL, Database Design & Optimization

Cloud & DevOps: Docker, Linux, AWS, Amazon EKS(Kubernetes), Microservices, Git, GitHub, CI/CD Pipelines, Postman, VS Code, Eclipse

System Design: AI Pipeline Architecture, Scalable System Design, Performance Optimization, Logging & Monitoring, Reliability Engineering

Soft Skills: Problem-Solving, Analytical Debugging, Teamwork, Technical Communication, Agile/Scrum Collaboration

KEY PROJECTS

AI Cricket Commentary App	2026
- Built and deployed an AI-powered cricket commentary application that generates real-time, natural-language match commentary using LLMs and prompt engineering. - Integrated Text-to-Speech (TTS) to convert generated commentary into natural-sounding audio output for a live, broadcast-style listening experience. - Designed a FastAPI backend to handle match-event data, trigger commentary generation, and serve responses with low latency. - Containerized the application with Docker and deployed it to the cloud, configuring environment variables and service endpoints for production use. - Tuned prompt templates and added output validation checks to reduce hallucinated or off-topic commentary. - Implemented caching and request batching to reduce LLM API costs while maintaining low-latency responses during live matches.	
LMS — Learning Management System	2025
- Developed and deployed a full-stack Learning Management System supporting course creation, content delivery, and student progress tracking. - Built RESTful APIs with Python/FastAPI and a React.js front end for course browsing, enrollment, and assessment workflows. - Designed the MySQL data model for courses, users, and progress records, and implemented role-based views for students and instructors. - Containerized the application with Docker and deployed it to the cloud, ensuring reliable access and straightforward release updates. - Implemented AI-assisted content recommendations to help students find relevant course materials based on their progress. - Wrote unit and integration tests for API endpoints, strengthening reliability and catching regressions early.	
Talent Management System — HRTech Platform	2025
- Built AI-assisted candidate profiling and resume-screening features using LLMs, integrated into the platform's backend services. - Designed and built RESTful APIs with Python and FastAPI to support interview scheduling and candidate-matching workflows. - Containerized services with Docker and deployed them to Amazon (EKS) for scalable, reliable operation. - Collaborated with cross-functional teams to integrate new AI-driven features into the platform's existing CI/CD pipelines. - Evaluated open-source and hosted LLM options to balance cost, latency, and output quality for resume-screening features. - Implemented fallback and retry logic to handle LLM provider rate limits and outages without disrupting the hiring workflow.	

EDUCATION

Bachelor of Engineering (B.E.) in Mechatronics — Sri Muthukumaran Institute of Technology, Chennai	Jun 2022 – Jun 2025
CGPA: 7.9 / 10	
Diploma in Mechanical Engineering (D.M.E.) — Central Polytechnic College, Chennai	Jul 2016 – May 2019
CGPA: 7.4 / 10	

CERTIFICATION

Java Full Stack Development — SLA Institute IBM Certified	May 2025
AI and full-stack development certification covering Java, Spring Boot, Hibernate, REST APIs, MySQL, and CI/CD, with hands-on capstone projects.	